Abstract

This study investigates the boundary constraints affecting the spreading kinematics of decane droplets on an aqueous cetyltrimethylammonium bromide (CTAB) surfactant bath. Experimental measurements of these interfacial flows can be highly sensitive to potential boundary interferences. Specifically, the goal of this work is to isolate the purely interfacial forces from potential edge effects, such as the pinning of the aqueous-plastic-air contact line at the container wall as well as viscous wall dissipation from shallow fluid depths. Preliminary testing was done to establish a repeatable methodology using angled high speed video to record the transition from the transient spreading dynamics across different solution concentrations. My initial parameter sweeps tested 52 mm, 85 mm, and 135 mm diameter containment vessels at a 5 mm substrate depth. Frame by frame processing in ImageJ revealed that standard 52 mm and 85 mm dishes prematurely lower the spreading rate due to edge effects. Conversely, the bigger 135 mm dish mitigated horizontal boundary interferences by providing a more sufficient unconstrained domain. Defining and eliminating these physical constraints is critical as it limits how thin a substrate layer can be generated without obscuring the fundamental fluid dynamics under investigation.

Introduction

This research is motivated by the need to accurately characterize capillary driven flows, specifically the spreading characteristics of decane droplets on a cetyltrimethylammonium bromide (CTAB) substrate interface. Interfacial spreading dynamics of this type are relevant to applications ranging from oil spill transport [11] to microfluidic and biological transport systems [7]. The measurement of these systems tend to be highly sensitive to physical boundary conditions. Currently, there is a lack of validation regarding how the actual physical constraints of the containment vessel influence the symmetry and velocity of the spreading front. There are two primary constraints that are the subject of this investigation: the thickness of the substrate fluid body and the inherent hydrophobicity of the containment dish. Edge effects and pinning are potential causes of inaccurate data, which could lead to a setup that does not adequately isolate the capillary and solutal forces at study as it limits how thin a layer can be generated.

The hypothesis of this research project in particular contains two parts. First, that if the CTAB substrate thickness is too shallow, the surface flow experiences viscous effects from the bottom surface of the container which can prematurely lower the spreading rate. Second, applying a non-chemical physical surface treatment to the walls of the containment vessel will decrease its inherent hydrophobicity, helping eliminate pinning effects without introducing chemical contaminants that could alter the interfacial stresses.

A baseline methodology has been established using various sizes of standard plastic petri dishes to isolate these horizontal and vertical interferences. To quantify the horizontal boundary constraints, my initial parameter sweeps evaluated decane spreading across 52 mm, 85mm, and 135mm diameter vessels at a constant 5 mm depth. This involved using high speed angled video to capture the transient spreading dynamics, with subsequent frame by frame processing in ImageJ to extract exact kinematics. By increasing the vessel diameter, this research aims to identify the critical distance at which the spreading front is no longer influenced by the pinned aqueous-plastic-air contact line at the wall. Furthermore, future iterations of this study will

utilize plasma treated glass petri dishes to modify surface wetting, which would allow for much lower substrate thicknesses to be tested than are possible in hydrophobic plastic, therefore allowing for the isolation of viscous wall dissipation from the bottom of the container.

Background

The spreading of an oil droplet on an aqueous surfactant solution typically follows a power-law relationship between the instantaneous droplet radius *R* and time *t* of the form [2]:

$$R \sim t^n$$

Understanding the transition from the Laplace to the Marangoni regime is foundational to this study. There are two distinct regimes that arise depending on the concentration of surfactant relative to the critical micelle concentration (CMC). When the concentration of surfactant is below the CMC (sub-CMC), spreading is driven by the Laplace pressure resulting from the mean curvature of the oil drop [1]. The balance between viscous forces and the surface tension gives the classical Laplace scaling law [1]:

$$R \sim \left(\frac{\gamma_{oa} V^2 l}{\mu}\right)^{1/8} t^{1/8},$$

where γ_{oa} is the oil-air interfacial tension, μ is the dynamic viscosity of the substrate liquid on which the oil drop spreads, and l is the depth of the substrate fluid layer. Notably, the depth *l* appears in this expression as viscous dissipation in the Laplace regime involves the full depth of the bath, where a shallower layer means greater viscous resistance and slower spreading [1, 4]. When the surfactant concentration is increased above the CMC (supra-CMC), the surface tension gradients drive a much faster Marangoni flow, where the power law takes the form:

$$R \sim \left(\frac{2S}{3\sqrt{\mu\rho}}\right)^{1/2} t^{3/4},$$

where S is the spreading coefficient [8] and ρ is the density of the substrate liquid. The spreading is significantly faster in this Marangoni regime, scaling with an exponent of $n = 3/4$ assuming a time independent spreading coefficient and up to $n = 5/4$ when accounting for adsorption-desorption dynamics [1, 4, 6]. Deodhar et al. mentioned that this transition is continuous and can be heavily dependent on the availability of surfactant molecules for bulk-interface exchange[1]. This theoretical scaling assumes an unconstrained interfacial domain. Experimental validation is highly sensitive to the physical constraints of the containment vessel, where in the case of our standard polystyrene petri dishes, there is a predominantly hydrophobic boundary.

There are two distinct sources of measurement error that motivate this work. The first is in the lateral direction, where as the spreading front approaches the vessel wall, it comes into interaction with the three-phase contact line formed between the aqueous solution, the container wall, and the air around. Contact angle hysteresis, where the contact line has a tendency to resist motion on surfaces due to energy barriers at surface defects, can cause this line to pin instead of receding freely, producing a retarding force on the approaching spreading front that isn't modeled in theory [3, 5]. On a hydrophobic polystyrene surface, this pinning is hypothesized to cause premature deceleration of the spreading front when it approaches the vessel wall, an effect

potentially significant in smaller diameter vessels when the front reaches the boundary sooner [5, 10]. The second error source is in the vertical direction, where a substrate thickness l that is too shallow can cause the momentum boundary layer to reach the container floor and disturb the power law behavior in both regimes, as is reflected in the laws above [1, 4]. The characterization of these container effects has not been reported significantly in literature, revealing a knowledge gap this work can address. Plasma surface treatment of glass vessels is a potential method of minimizing wall pinning without introducing chemical contaminants, enabling experiments at substrate depths currently inaccessible with untreated polystyrene [5, 10].

Testing these boundary hypotheses will play an important role in ensuring that future lab results are not skewed by the different parameters at play, ultimately leading to the development of a more robust setup for studying related viscous flows.

Preliminary results

Scope and Purpose

The goal of this preliminary investigation is to validate the existing experimental setup by quantifying the impact of the vessel diameter and surface wetting on decane spreading kinematics. The current phase of the project focuses on establishing baseline kinetics across different container diameters and surfactant concentrations to verify that interfacial forces can be successfully isolated.

Methodology

To evaluate the impact of the spatial boundary constraints on spreading kinematics, baseline tests were conducted using standard polystyrene petri dishes. Aqueous CTAB solutions were prepared at 0.5 mM (sub-CMC) and 2.0 mM (supra-CMC) concentrations and poured into the containment vessels and allowed to reach equilibrium. Initial baseline trials involved a four hour equilibrium period to ensure complete stabilization of the surfactant body. However, subsequent control tests demonstrated that this extended timeframe was unnecessary for the concentrations evaluated, and following trials omitted this step. Decane droplets were dispensed at the center point of the substrate, and the transient spreading phases were recorded with a high speed video setup capturing at rates ranging from 60 to 600 frames per second [9]. The resulting video files were each analyzed frame-by-frame using ImageJ to extract the exact spreading radius $R(t)$.

To evaluate the horizontal boundary interference, a spatial parameter sweep was conducted. The 0.5 mM and 2.0 mM solutions were tested across three different vessel diameters: 52 mm, 85 mm, and 135 mm, while holding the substrate thickness constant at 5mm to serve as a control. This isolated the lateral confinement variable, evaluating whether the standard vessel boundaries are sufficient to alter the spreading kinematics through interaction with the pinned wall meniscus.

To quantitatively compare the spreading rates, the extracted data was modeled using a power-law relationship, $\boldsymbol{R = At^n}$. By fixing the scaling exponent n to the theoretical constants associated

with the specific flow regime, the pre-factor A was empirically calculated to serve as a quantifiable index of spreading velocity.

Results

Table 1: Summary of Experiment Trials

Concentration	Vessel Diameter	Frame Rate (fps)	Burst Frame
0.5 mM	52 mm	60	Frame 10
0.5 mM	85 mm	60	Frame 97 / Frame 77 (T1 / T2)
0.5 mM	135 mm	60	Frame 56
2.0 mM	52 mm	600	Frame 93
2.0 mM	85 mm	600	Frame 274 / Frame 67 (T1 / T2)
2.0 mM	135 mm	600	Frame 196

Taken with constant 5 mm substrate depth and polystyrene petri dishes

The baseline spreading tests confirm the expected distinction between sub-CMC and supra-CMC regimes, as shown in Figure 1. The 2.0 mM supra-CMC trials reached a maximum spreading radius of approximately 30 mm within 0.37 seconds, while the 0.5 mM sub-CMC trials required

around 6 seconds to reach a comparable extent of 22mm, where there is a large difference in timescale consistent with the transition from Laplace to Marangoni dominated spreading [1].

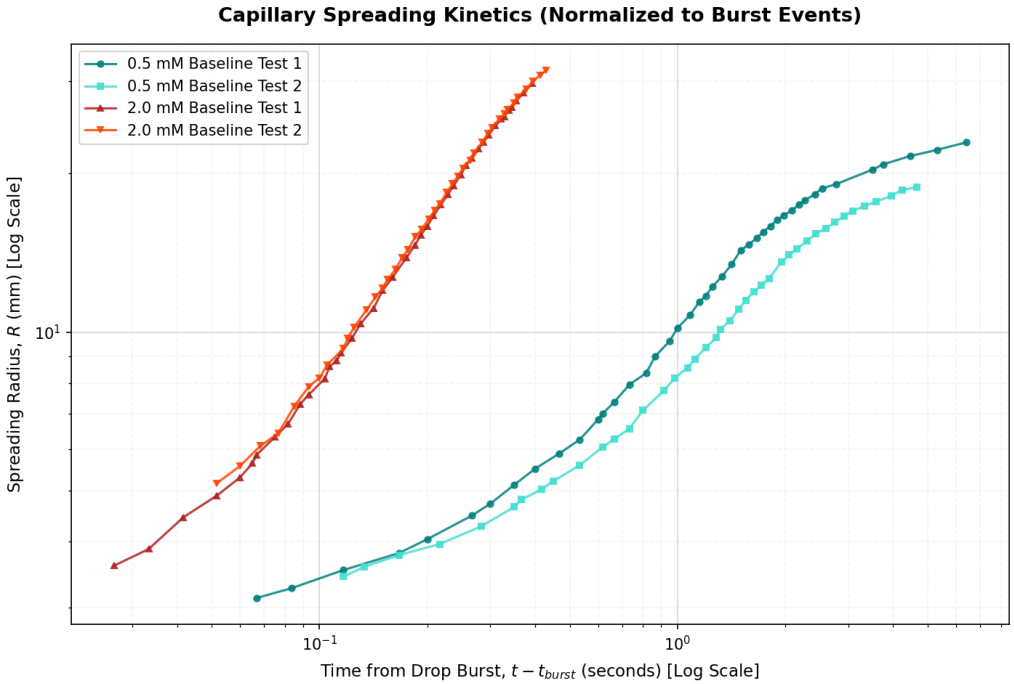


Figure 1: Log-log plot illustrating surfactant-driven capillary spreading kinetics of decane droplet on 0.5 mM sub-CMC and 2.0 mM supra-CMC environments. Trials conducted in standard 85mm diameter containment vessels at constant 5 mm fluid depth.

To further evaluate the impact of lateral confinement, the geometric sweep data for both the 0.5 mM and 2.0 mM concentrations were plotted to visualize the onset of boundary-induced deceleration (Figures 2 and 3)

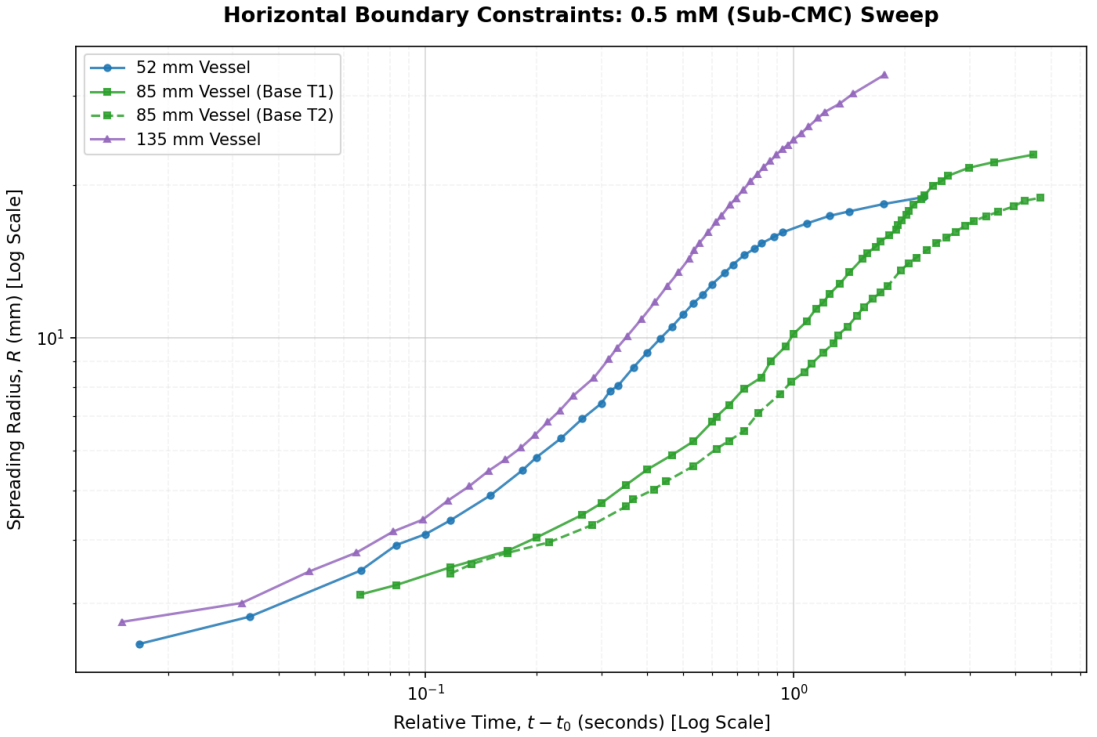


Figure 2: Comparison of decane spreading in 0.5 mM CTAB solution for different vessel diameters plotted on log-log scale

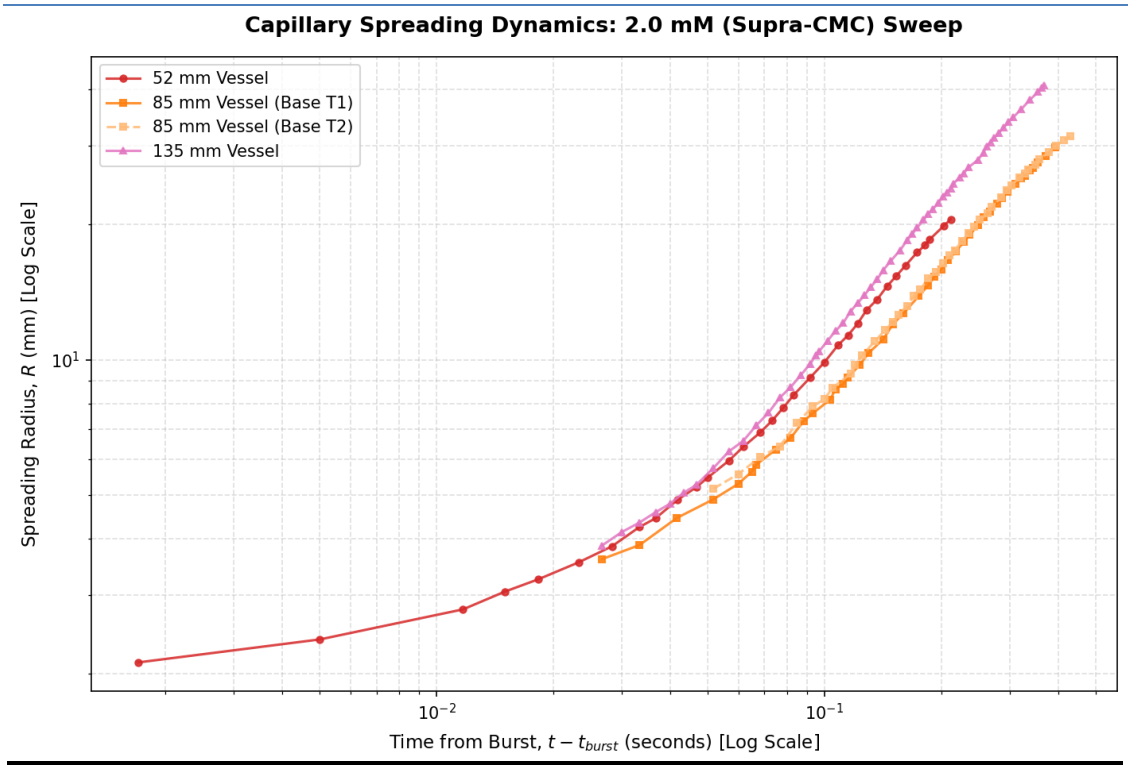


Figure 3: Comparison of decane spreading in 2.0 mM CTAB solution for different vessel diameters plotted on log-log scale

Discussion

The most notable finding from this geometric sweep is that during the primary stages of spreading, the curves for all dish sizes line up closely. This is a crucial result because it proves that the baseline chemistry and the raw driving forces were completely consistent across every single run. The container size does not change how fast the droplet wants to expand at the start. Instead, the differences primarily show up in the extreme case. The 52 mm dish curve shows a clear divergence point where it flattens out early. While we could not directly measure the wall profile with our top-down camera setup, this premature plateauing matches exactly what you would expect from severe edge constraints. As the fluid front gets too close to the 52 mm walls, the pinned meniscus likely creates a physical backpressure that fights the outward flow and forces it to a premature stop.

When comparing our standard 85 mm baseline to the larger 135 mm dish, however, the benefit is more subtle. The two datasets track very closely together throughout the observation window, meaning that the standard 85 mm dish is largely sufficient for capturing the core power-law kinetics without immediate boundary interference. The 135 mm dish does offer a clear improvement by moving the walls further away and maximizing the available runway, but it serves more as an optimized safety margin to ensure unhindered expansion rather than a complete departure from the 85 mm performance. This indicates that while the 52 mm size is

clearly insufficient, the standard 85 mm dish remains a highly reliable benchmark for general testing, with the 135 mm dish providing the ultimate unconstrained continuum limit.

Conclusion

This study successfully validated a repeatable experimental protocol for capturing Marangoni and capillary spreading phases across sub-CMC and supra-CMC concentration regimes using high-speed video and frame-by-frame ImageJ analysis. Baseline spreading tests confirm the expected qualitative and quantitative distinction between the two regimes, with supra-CMC Marangoni spreading occurring approximately ten times faster than sub-CMC capillary spreading. Forced power-law fitting of the dominant regime yields a 2.0 mM prefactor of 93.1, in close agreement with an established laboratory baseline value of 97.0, providing robust validation of the measurement methodology.

The horizontal boundary parameter sweep provides preliminary results demonstrating that vessel diameter directly influences the observable spreading kinematics. The 52 mm dish, which forces the droplet to reach the largest fraction of its inner radius (79% for 2.0 mM, 73% for 0.5 mM), exhibits a clearly constrained spreading domain characterized by early kinetic plateauing. However, comparisons between the 85 mm and 135 mm vessels indicate that while the 135 mm dish provides the maximum unconstrained tracking window, the standard 85 mm baseline remains highly reliable for capturing core power-law kinetics before severe boundary interaction occurs.

Because these findings represent the preliminary phase of an ongoing investigation, the quantitative differences in pre-factors across dish sizes require further statistical validation. The next phase of this study will conduct replicate trials across all vessel sizes, transition to plasma-treated glass petri dishes to eliminate contact angle hysteresis as a variable and conduct a systematic substrate depth sweep to isolate viscous wall dissipation effects [5, 10]. Together, these refinements will build upon these preliminary results to produce a fully optimized experimental platform for quantitatively studying the boundary-dependence of interfacial spreading.

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